

Complete JavaScript Roadmap — Topics + Tasks + Projects

Phase 0 — JavaScript Introduction

Topics

- What is JavaScript?
- JavaScript vs HTML vs CSS
- JavaScript engines
- Runtime environment
- Browser JavaScript
- Node.js JavaScript
- How JavaScript code executes
- `<script>` tag
- External JS files
- "use strict"
- Comments
- Statements
- Expressions

Tasks

- Write your first JS program
- Connect JS to an HTML file
- Use inline, internal, and external JavaScript
- Print different values using `console.log()`
- Identify statements vs expressions
- Test strict mode

Project

- HTML page with buttons
 - Each button performs a different JS action
 - Console output + visible output
-

Phase 1 — Variables

Topics

- Variables
- var
- let
- const
- Declaration
- Initialization
- Reassignment
- Naming rules
- Scope basics

Tasks

- Create variables of different types
- Reassign let
- Try reassigning const
- Compare var, let, and const
- Create variables using meaningful names
- Build calculations using variables

Project

Personal Profile Generator

- Name
- Age
- City
- Education
- Skills
- Display everything dynamically

Phase 2 — Data Types

Topics

- String
- Number
- BigInt
- Boolean
- Undefined
- Null
- Symbol
- Object
- Array
- Function
- typeof
- Primitive vs reference values
- Mutable vs immutable values

Tasks

- Identify the type of 20 different values
- Convert values between types
- Test typeof
- Compare primitive and reference values
- Experiment with null and undefined
- Create mixed arrays and objects

Project

Student Information System

- Store student information
- Display each property
- Check the data type of every value

Phase 3 — Operators

Topics

- Arithmetic operators
- Assignment operators
- Comparison operators
- Logical operators
- Ternary operator
- ??
- ?.
- typeof
- instanceof
- in
- Operator precedence
- Short-circuit evaluation

Tasks

- Create arithmetic calculators
- Compare numbers
- Create logical conditions
- Practice &&, ||, !
- Practice ternary operators
- Practice optional chaining
- Predict operator results

Project

Smart Calculator

- Addition
- Subtraction
- Multiplication
- Division
- Percentage

- Comparison
 - Error handling
-

Phase 4 — Type Conversion & Coercion

Topics

- Implicit conversion
- Explicit conversion
- String()
- Number()
- Boolean()
- parseInt()
- parseFloat()
- toString()
- Truthy
- Falsy
- NaN
- Infinity
- Number.isNaN()
- Number.isFinite()
- == vs ===

Tasks

- Convert strings to numbers
- Convert numbers to strings
- Test truthy/falsy values
- Predict coercion results
- Debug unexpected conversions
- Create type-checking functions

Project

Input Converter

- User enters any value
 - Show original value
 - Show string version
 - Show number version
 - Show boolean version
-

Phase 5 — Strings

Topics

- String creation
- Indexing
- Length
- Escape characters
- Template literals
- Concatenation
- toUpperCase()
- toLowerCase()
- trim()
- includes()
- startsWith()
- endsWith()
- indexOf()
- slice()
- substring()
- replace()
- replaceAll()
- split()
- charAt()

- `at()`
- `repeat()`

Tasks

- Reverse a string
- Count characters
- Count words
- Find a character
- Replace words
- Check palindrome
- Capitalize names
- Extract parts of strings

Project

Text Analyzer

- Character count
- Word count
- Uppercase/lowercase
- Search word
- Replace word
- Reverse text

Phase 6 — Conditional Logic

Topics

- `if`
- `else`
- `else if`
- Nested conditions
- Ternary
- `switch`

- case
- break
- default
- Guard clauses

Tasks

- Even/odd checker
- Age checker
- Grade calculator
- Login validation
- Positive/negative checker
- Largest of three numbers
- Traffic-light logic

Project

Student Grade System

- Enter marks
- Calculate percentage
- Assign grade
- Pass/fail
- Remarks

Phase 7 — Loops

Topics

- for
- while
- do...while
- Nested loops
- break
- continue

- for...of
- for...in
- forEach()

Tasks

- Print numbers
- Print even numbers
- Print odd numbers
- Multiplication tables
- Sum numbers
- Find factorial
- Reverse numbers
- Nested pattern problems
- Loop through arrays
- Loop through objects

Project

Number Analyzer

- Generate numbers
- Even/odd
- Prime numbers
- Sum
- Average
- Maximum
- Minimum

Phase 8 — Functions

Topics

- Function declaration
- Function expression

- Parameters
- Arguments
- Return
- Default parameters
- Rest parameters
- Nested functions
- Callback functions
- Higher-order functions
- Pure functions
- Side effects
- Arrow functions

Tasks

- Build calculator functions
- Create reusable validation functions
- Create functions with multiple parameters
- Return objects from functions
- Create callback functions
- Convert normal functions to arrow functions

Project

Utility Function Library

Create reusable functions for:

- Calculator
- Validation
- String manipulation
- Number manipulation
- Date formatting

Phase 9 — Arrays

Topics

- Creating arrays
- Indexing
- Length
- Updating elements
- Nested arrays
- push()
- pop()
- shift()
- unshift()
- slice()
- splice()
- concat()
- join()
- includes()
- indexOf()
- reverse()
- sort()
- flat()
- flatMap()
- forEach()
- map()
- filter()
- find()
- findIndex()
- some()
- every()
- reduce()

Tasks

- Add/remove items
- Search array
- Sort array
- Reverse array
- Find maximum
- Find minimum
- Filter data
- Transform data
- Calculate totals
- Group/filter information

Project

Todo List — Console Version

Features:

- Add todo
- Delete todo
- Complete todo
- Search todo
- Filter completed todos
- Count todos

Phase 10 — Objects

Topics

- Object literals
- Properties
- Methods
- Dot notation
- Bracket notation
- Adding properties

- Updating properties
- Deleting properties
- Nested objects
- Objects inside arrays
- Arrays inside objects
- Dynamic properties
- Object.keys()
- Object.values()
- Object.entries()
- Object.assign()
- Object.freeze()
- Object.seal()

Tasks

- Create user objects
- Update properties
- Add/delete properties
- Create nested objects
- Loop through objects
- Convert object data
- Search objects

Project

Student Management System

- Add students
 - Store student data
 - Search student
 - Update student
 - Delete student
 - Calculate grades
-

Phase 11 — Destructuring & Spread

Topics

- Array destructuring
- Object destructuring
- Default values
- Renaming
- Nested destructuring
- Function parameter destructuring
- Array spread
- Object spread
- Rest parameters
- Rest destructuring

Tasks

- Extract values from arrays
- Extract values from objects
- Rename properties
- Copy arrays
- Copy objects
- Merge arrays
- Merge objects
- Remove selected properties

Project

User Data Transformer

Take complex user objects and produce:

- Profile object
- Contact object
- Account object
- Display object

Phase 12 — Scope

Topics

- Global scope
- Function scope
- Block scope
- Lexical scope
- Scope chain
- Variable shadowing
- var
- let
- const

Tasks

- Predict variable accessibility
- Fix scope errors
- Create nested scopes
- Compare var and let
- Identify shadowing

Project

Scope Challenge

Create 15 deliberately tricky scope problems and solve/debug each one.

Phase 13 — Hoisting

Topics

- Variable hoisting
- Function hoisting
- var
- let

- `const`
- Temporal Dead Zone

Tasks

- Predict outputs before running code
- Debug hoisting errors
- Compare declaration behavior
- Explain why each result occurs

Project

JavaScript Execution Quiz

Create a quiz where users predict the output of tricky JS snippets.

Phase 14 — JavaScript Execution

Topics

- Execution context
- Global execution context
- Function execution context
- Call stack
- Memory allocation
- Creation phase
- Execution phase
- Scope chain
- Lexical environment
- Garbage collection

Tasks

- Draw execution-context diagrams
- Trace function calls
- Trace the call stack
- Predict execution order

- Identify memory/reference behavior

Project

Call Stack Visualizer

Build a simple visual representation of functions entering/leaving the call stack.

Phase 15 — Closures

Topics

- Closures
- Lexical closures
- Function returning function
- Private data
- Closures in loops
- Closures in callbacks

Tasks

- Create counter closure
- Create private variables
- Create function factories
- Solve closure loop problems

Project

Private Counter System

Create:

- Increment
- Decrement
- Reset
- Get value

The internal value should not be directly accessible.

Phase 16 — this

Topics

- Global this
- Object this
- Function this
- Arrow-function this
- Event-handler this
- call()
- apply()
- bind()

Tasks

- Predict this
- Fix broken methods
- Compare arrow vs normal functions
- Practice call
- Practice apply
- Practice bind

Project

Bank Account Object

Implement:

- Deposit
- Withdraw
- Balance
- User information
- Different methods using this

Phase 17 — DOM

Topics

- DOM

- DOM tree
- Element selection
- `querySelector()`
- `querySelectorAll()`
- `textContent`
- `innerHTML`
- Attributes
- `classList`
- Styles
- Creating elements
- Removing elements
- Replacing elements

Tasks

- Select elements
- Change text
- Change styles
- Add classes
- Create elements
- Delete elements
- Generate HTML dynamically

Project

Dynamic Todo App

Features:

- Add
- Delete
- Complete
- Edit
- Filter
- Counter

- Empty state
-

Phase 18 — Events

Topics

- Events
- `addEventListener()`
- Click
- Input
- Change
- Submit
- Keyboard
- Mouse
- Event object
- `target`
- `currentTarget`
- Bubbling
- Capturing
- Propagation
- `preventDefault()`
- Event delegation

Tasks

- Build click counters
- Keyboard shortcuts
- Dynamic button events
- Form submission
- Event delegation
- Prevent default behavior

Project

Interactive Dashboard

Include:

- Buttons
 - Keyboard shortcuts
 - Tabs
 - Dropdown
 - Modal
 - Notifications
 - Dynamic content
-

Phase 19 — Forms

Topics

- Form elements
- Reading values
- Form submission
- Validation
- Required fields
- Custom validation
- Error messages
- Form state

Tasks

- Validate username
- Validate email
- Validate password
- Confirm password
- Show errors
- Clear errors

Project

Registration Form

Features:

- Name
 - Email
 - Password
 - Confirm password
 - Validation
 - Error messages
 - Success screen
-

Phase 20 — Browser APIs

Topics

- window
- document
- console
- alert()
- confirm()
- prompt()
- location
- history
- navigator
- localStorage
- sessionStorage
- JSON

Tasks

- Save data in localStorage
- Read saved data
- Delete saved data

- Create theme persistence
- Store user preferences

Project

Persistent Notes App

Features:

- Create notes
 - Edit notes
 - Delete notes
 - Search
 - localStorage
 - Dark/light mode
-

Phase 21 — Date & Time

Topics

- Date
- Timestamps
- Date.now()
- Getting dates
- Setting dates
- Formatting
- Comparing dates
- Time zones
- Intl.DateTimeFormat

Tasks

- Digital clock
- Countdown
- Age calculator
- Date difference

- Format dates

Project

Countdown Application

- Target date
 - Days
 - Hours
 - Minutes
 - Seconds
 - Completed state
-

Phase 22 — Math & Numbers

Topics

- Math
- Rounding
- Floor
- Ceiling
- Random numbers
- Min/max
- Absolute values
- Powers
- Number formatting
- Decimal precision

Tasks

- Random number generator
- Dice simulator
- Password number generator
- Percentage calculator
- Random color generator

Project

Number Guessing Game

Phase 23 — Regular Expressions

Topics

- Regex syntax
- Character classes
- Quantifiers
- Anchors
- Groups
- Alternation
- Flags
- `test()`
- `match()`
- `replace()`

Tasks

- Validate email
- Validate phone
- Validate username
- Extract numbers
- Find words
- Replace patterns

Project

Advanced Form Validator

Validate:

- Email
- Pakistani phone number
- Username

- Password
 - CNIC format
 - Website URL
-

Phase 24 — JSON

Topics

- JSON structure
- JSON vs JavaScript objects
- JSON.parse()
- JSON.stringify()
- Nested JSON
- JSON APIs
- Sending JSON
- Receiving JSON

Tasks

- Convert object → JSON
- Convert JSON → object
- Store JSON in localStorage
- Read JSON data
- Manipulate nested JSON

Project

JSON Data Explorer

Load a JSON dataset and allow users to:

- Search
- Filter
- Sort
- View
- Count records

Phase 25 — Asynchronous JavaScript

Topics

- Synchronous vs asynchronous
- Blocking vs non-blocking
- Callbacks
- Callback hell
- Promises
- Promise states
- .then()
- .catch()
- .finally()
- Promise chaining
- Promise.all()
- Promise.allSettled()
- Promise.race()
- Promise.any()

Tasks

- Create delayed functions
- Create custom promises
- Chain promises
- Handle errors
- Run multiple promises
- Compare promise methods

Project

Fake API System

Simulate:

- User loading

- Products loading
 - Orders loading
 - Errors
 - Loading states
-

Phase 26 — Async/Await

Topics

- async
- await
- Async functions
- try...catch
- Sequential operations
- Parallel operations

Tasks

- Convert promises to async/await
- Handle errors
- Run parallel requests
- Create loading states

Project

Weather Dashboard

- Search city
 - Fetch data
 - Loading state
 - Error state
 - Display weather
-

Phase 27 — Event Loop

Topics

- Call stack
- Web APIs
- Callback queue
- Microtask queue
- Macrotasks
- Event loop
- Promise execution order
- setTimeout()
- setInterval()
- setImmediate()
- process.nextTick()

Tasks

- Predict execution order
- Analyze mixed promises/timeouts
- Trace the event loop
- Create event-loop diagrams

Project

Event Loop Simulator

Display:

- Call stack
- Microtask queue
- Callback queue
- Execution order

Phase 28 — Fetch & APIs

Topics

- HTTP basics
- Requests

- Responses
- GET
- POST
- PUT
- PATCH
- DELETE
- fetch()
- Headers
- Request body
- Status codes
- response.json()
- API errors
- Loading states

Tasks

- GET API data
- POST data
- Update data
- Delete data
- Handle errors
- Handle loading states

Project

CRUD API Frontend

Build a frontend that can:

- Create
- Read
- Update
- Delete
- Search
- Filter

Phase 29 — Error Handling

Topics

- Syntax errors
- Runtime errors
- Logical errors
- try
- catch
- finally
- throw
- Custom errors
- Error
- TypeError
- ReferenceError

Tasks

- Intentionally create errors
- Read stack traces
- Fix runtime errors
- Create custom errors
- Handle API errors

Project

Error Handling Playground

Give users broken JavaScript examples and let them identify/fix the problem.

Phase 30 — Modules

Topics

- Modules
- ES Modules

- export
- import
- Named exports
- Default exports
- Import aliases
- Dynamic imports
- CommonJS
- require()
- module.exports

Tasks

- Split code into modules
- Import functions
- Export objects
- Create reusable utilities
- Compare CommonJS and ES Modules

Project

Modular Expense Tracker

Separate:

- UI
- Calculations
- Storage
- Validation
- Data management

Phase 31 — Object-Oriented JavaScript

Topics

- Constructor functions
- Prototypes

- Prototype chain
- Classes
- Constructor
- Methods
- Instance properties
- Static methods
- Getters
- Setters
- Inheritance
- extends
- super
- Encapsulation
- Polymorphism

Tasks

- Create classes
- Create instances
- Add methods
- Use inheritance
- Create static methods
- Use getters/setters
- Explore prototypes

Project

Banking System

Classes:

- User
- Account
- SavingsAccount
- Transaction

Features:

- Deposit
 - Withdraw
 - Transfer
 - Transaction history
-

Phase 32 — Advanced Functions

Topics

- First-class functions
- Higher-order functions
- Closures
- Currying
- Partial application
- Function composition
- IIFE
- Memoization
- Recursion
- Debouncing
- Throttling

Tasks

- Recursive factorial
- Recursive Fibonacci
- Function factory
- Memoized function
- Debounced search
- Throttled scroll

Project

Search Engine Interface

Include:

- Search
 - Debounced input
 - Filtering
 - Sorting
 - Result count
 - Search history
-

Phase 33 — Advanced Objects

Topics

- Property descriptors
- Getters/setters
- Prototype inheritance
- Object.create()
- Object.defineProperty()
- Symbols
- Iterators
- Generators
- Proxy
- Reflect

Tasks

- Create custom properties
- Build an iterator
- Build a generator
- Use Proxy
- Intercept object operations
- Explore property descriptors

Project

Reactive Object System

Build a small system that detects when object properties change and updates the UI.

Phase 34 — Advanced Data Structures

Topics

- Set
- WeakSet
- Map
- WeakMap
- Typed Arrays
- ArrayBuffer
- DataView

Tasks

- Remove duplicates with Set
- Create key-value systems with Map
- Compare Map vs Object
- Practice WeakMap
- Work with binary data

Project

Advanced Data Manager

Build a data-management system using:

- Map
- Set
- WeakMap
- Arrays
- Objects

Phase 35 — Modern JavaScript

Topics

- ES6+
- Modern syntax
- Modern array methods
- Modern object methods
- Optional chaining
- Nullish coalescing
- Private class fields
- Dynamic imports
- Modern modules

Tasks

- Refactor old JS code
- Convert old syntax to modern syntax
- Use modern array/object methods
- Replace unnecessary conditionals

Project

Legacy JS → Modern JS Refactor

Take an old JavaScript project and completely modernize its code.

Phase 36 — Debugging

Topics

- Reading error messages
- Syntax debugging
- Logic debugging
- Runtime debugging
- `console.log()`
- `console.table()`
- Breakpoints
- Step over

- Step into
- Step out
- Watch variables
- Call stack
- Network tab
- Sources tab
- Application tab
- DOM inspection
- Root-cause analysis
- Async debugging

Tasks

- Fix broken code
- Find logic bugs
- Debug DOM problems
- Debug API requests
- Debug asynchronous code
- Trace call stacks

Project

Broken App Challenge

Take a deliberately broken application and fix:

- 20+ bugs
- UI bugs
- Logic bugs
- DOM bugs
- Async bugs
- API bugs

Topics

- XSS
- DOM XSS
- Input sanitization
- innerHTML risks
- textContent
- CORS
- Same-origin policy
- Client-side validation
- Server-side validation
- Client data trust

Tasks

- Identify XSS vulnerabilities
- Replace unsafe DOM manipulation
- Test malicious inputs
- Understand CORS errors
- Compare frontend/backend validation

Project

Secure Login Demo

Build a basic login/register interface and identify/fix common client-side security problems.

Phase 38 — JavaScript Performance

Topics

- DOM performance
- Event delegation
- Debouncing
- Throttling
- Memory leaks

- Garbage collection
- Efficient loops
- Array performance
- Rendering performance
- Lazy loading
- Code splitting

Tasks

- Optimize slow loops
- Optimize DOM manipulation
- Compare different array methods
- Implement debounce
- Implement throttle
- Find memory leaks

Project

Performance Challenge

Create a slow application, measure it, then optimize it and compare the results.

Phase 39 — Node.js Preparation

Topics

- Node.js runtime
- Browser vs Node.js
- Global objects
- process
- Environment variables
- File system basics
- Modules
- npm
- package.json

- npm scripts
- Dependencies
- Dev dependencies
- CommonJS
- ES Modules
- Async programming
- Event loop
- HTTP basics
- REST API concepts
- JSON
- Error handling

Tasks

- Create Node project
- Initialize npm
- Install packages
- Create npm scripts
- Read environment variables
- Read/write files
- Create modules
- Create basic HTTP server

Project

Node.js CLI + API

Build a small backend-ready application with:

- Node.js
- npm
- Modules
- Environment variables
- File system
- HTTP server

- JSON
 - Error handling
-



Final JavaScript Projects

After completing the roadmap, don't immediately jump into another tutorial.

Build these **from scratch**:

Project 1 — Todo Application

HTML + CSS + JS + DOM + localStorage

Project 2 — Expense Tracker

Arrays + objects + functions + localStorage

Project 3 — Quiz Application

DOM + events + arrays + objects + timers

Project 4 — Weather Dashboard

Async JS + Fetch + API + error handling

Project 5 — Movie Search App

API + async/await + search + debounce

Project 6 — E-Commerce Frontend

Products + filtering + cart + localStorage

Project 7 — CRUD Application

Frontend + API + GET/POST/PUT/DELETE

Project 8 — Admin Dashboard

DOM + APIs + charts + authentication concepts

Project 9 — Full JavaScript Application

Modules + APIs + async + localStorage + advanced JS

Project 10 — Final Challenge

Build a complete application **without following a tutorial**, using only documentation when necessary.